

Product Review and Rating Management System

Snapdeal_e-commerce_project

1.Create Rating Table

Attribute	Key Type	Data Type	Size
Rating_ID	Primary Key	NUMBER	5
Product_ID	Foreign Key	NUMBER	5
Customer_ID	Foreign Key	NUMBER	5
Rating	-	NUMBER	1
Rating_Date	-	DATE	-

```
SQL> CREATE TABLE Rating (  
2     Rating_ID NUMBER(5) PRIMARY KEY,  
3     Product_ID NUMBER(5),  
4     Customer_ID NUMBER(5),  
5     Rating NUMBER(1),  
6     Rating_Date DATE,  
7     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID),  
8     FOREIGN KEY (Customer_ID) REFERENCES Customer(ID),  
9     CHECK (Rating BETWEEN 1 AND 5)  
10 );  
  
Table created.
```

2.Create Review Table :

Attribute	Key Type	Data Type	Size
Review_ID	Primary Key	NUMBER	5
Customer_ID	Foreign Key	NUMBER	5
Product_ID	Foreign Key	NUMBER	5
Review_Text	-	VARCHAR2	500
Review_Date	-	DATE	-

3.Insert Values for Rating Table:

```

SQL> INSERT INTO Rating VALUES
  2 (1, 23, 1, 5, DATE '2026-10-01');

1 row created.

SQL>
SQL> INSERT INTO Rating VALUES
  2 (2, 24, 2, 4, DATE '2026-10-02');

1 row created.

SQL>
SQL> INSERT INTO Rating VALUES
  2 (3, 25, 3, 5, DATE '2026-10-03');

1 row created.

SQL>
SQL> INSERT INTO Rating VALUES
  2 (4, 26, 4, 5, DATE '2026-10-04');

1 row created.

SQL>
SQL> INSERT INTO Rating VALUES
  2 (5, 27, 5, 4, DATE '2026-10-05');

1 row created.

```

4.Insert Values for Review Table:

```

SQL> INSERT INTO Review VALUES
  2 (1, 1, 23, 'Excellent running shoes and very comfortable', DATE '2026-10-01');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2 (2, 2, 24, 'Good quality casual shoes and nice design', DATE '2026-10-02');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2 (3, 3, 25, 'Very durable and comfortable sports shoes', DATE '2026-10-03');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2 (4, 4, 26, 'Very useful book for self improvement', DATE '2026-10-04');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2 (5, 5, 27, 'Good book for learning Python programming', DATE '2026-10-05');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2 (6, 6, 28, 'Interesting book about personal finance', DATE '2026-10-06');

1 row created.

SQL>
SQL> INSERT INTO Review VALUES
  2 (7, 1, 29, 'Good face cream with moisturizing effect', DATE '2026-10-07');

```

5.Retrieve product Review Details :

```
SQL> SELECT * FROM Review;

REVIEW_ID CUSTOMER_ID PRODUCT_ID
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
      1          1          23
Excellent running shoes and very comfortable
01-OCT-26

      2          2          24
Good quality casual shoes and nice design
02-OCT-26

      3          3          25
Very durable and comfortable sports shoes
03-OCT-26

      4          4          26
Very useful book for self improvement
```

6.Retrieve Product Rating Details :

```
SQL> SELECT * FROM Rating;

RATING_ID PRODUCT_ID CUSTOMER_ID    RATING RATING_DA
-----
      1          23          1          5 01-OCT-26
      2          24          2          4 02-OCT-26
      3          25          3          5 03-OCT-26
      4          26          4          5 04-OCT-26
      5          27          5          4 05-OCT-26
      6          28          6          3 06-OCT-26
      7          29          1          4 07-OCT-26
      8          30          2          5 08-OCT-26
      9          31          3          4 08-OCT-26
     10          32          4          3 08-OCT-26
     11          33          5          4 08-OCT-26

RATING_ID PRODUCT_ID CUSTOMER_ID    RATING RATING_DA
-----
     12          34          6          5 08-OCT-26
     13          35          1          5 08-OCT-26
     14          36          2          4 08-OCT-26

14 rows selected
```

7.Retrieve Product Review Details with Product and Customer Names:

```
SQL> SELECT P.Product_ID,
2      P.Product_Name,
3      C.ID AS Customer_ID,
4      C.Name AS Customer_Name,
5      R.Review_Text,
6      R.Review_Date
7  FROM Product P, Customer C, Review R
8  WHERE P.Product_ID = R.Product_ID
9  AND C.ID = R.Customer_ID
10 ORDER BY P.Product_ID;
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
CUSTOMER_ID CUSTOMER_NAME
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
      23
Nike Running Shoes
      1 Ramesh

PRODUCT_ID
-----
PRODUCT_NAME
-----
CUSTOMER_ID CUSTOMER_NAME
-----
REVIEW_TEXT
-----
```

8. Calculate Average Product Ratings :

```
SQL> SELECT Product_ID,
2      AVG(Rating) AS Average_Rating
3  FROM Rating
4  GROUP BY Product_ID
5  HAVING AVG(Rating) >= 4
6  ORDER BY Average_Rating DESC;
```

```
PRODUCT_ID AVERAGE_RATING
-----
35          5
34          5
23          5
26          5
30          5
25          5
33          4
31          4
36          4
27          4
24          4

PRODUCT_ID AVERAGE_RATING
-----
29          4
```

12 rows selected.

```
SQL> SELECT Product_ID,
2      AVG(Rating) AS Average_Rating
3  FROM Rating
4  GROUP BY Product_ID
5  ORDER BY Product_ID;
```

PRODUCT_ID	AVERAGE_RATING
23	5
24	4
25	5
26	5
27	4
28	3
29	4
30	5
31	4
32	3
33	4

PRODUCT_ID	AVERAGE_RATING
34	5
35	5
36	4

14 rows selected.

9. Identify Highly Rated Products :

```
SQL> SELECT Product_ID,
2      AVG(Rating) AS Average_Rating
3  FROM Rating
4  GROUP BY Product_ID
5  HAVING AVG(Rating) >= 4
6  ORDER BY Average_Rating DESC;
```

PRODUCT_ID	AVERAGE_RATING
35	5
34	5
23	5
26	5
30	5
25	5
33	4
31	4
36	4
27	4
24	4

PRODUCT_ID	AVERAGE_RATING
29	4

12 rows selected.

10.Product Rating Report :

```
SQL> SELECT COUNT(Rating) AS Total_Ratings,  
2      SUM(Rating) AS Total_Rating,  
3      AVG(Rating) AS Average_Rating,  
4      MAX(Rating) AS Highest_Rating,  
5      MIN(Rating) AS Lowest_Rating  
6  FROM Rating;  
  
TOTAL_RATINGS  TOTAL_RATING  AVERAGE_RATING  HIGHEST_RATING  LOWEST_RATING  
-----  
14             60          4.28571429          5              3  
SQL>
```

11.Conclusion :

The Product Review and Rating Management System was successfully created using SQL. Review and Rating tables were created to store customer feedback and ratings for products. Customer reviews and ratings were inserted and retrieved successfully. Aggregate functions such as `AVG()`, `COUNT()`, `SUM()`, `MAX()`, and `MIN()` were used to analyze product ratings. Highly rated products were also identified based on their average ratings. Thus, the system helps in effectively managing and analyzing customer feedback and product ratings.